• NOISE—4X4—FRONT AXLE NOISE IN 2 HIGH ONLY • AXLE—4X4—FRONT AXLE NOISE IN 2 HIGH ONLY	Article No. 02-25-7

FORD: 2003 EXPEDITION

LINCOLN: 2003 NAVIGATOR

ISSUE

Some 2003 Expedition/Navigator 4x4 vehicles may exhibit a front axle groan or hum, and/or a slight vibration, when operated in the 2 High mode. This condition is most noticeable on deceleration between 45 and 15 mph, and is not present in A4WD or 4 High. This may be due to the front halfshafts rotating at different speeds when the vehicle is operated in 2 High.

ACTION

To service, install S-spring (clutch spring) into the front axle differential, causing the front halfshafts to rotate at the same speed in 2 High mode. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

WARNING

THE ELECTRICAL POWER TO THE AIR SUSPENSION SYSTEM MUST BE SHUT OFF PRIOR TO HOISTING, JACKING OR TOWING AN AIR SUSPENSION VEHICLE. THIS CAN BE ACCOMPLISHED BY TURNING OFF THE AIR SUSPENSION SWITCH LOCATED IN THE LEFT REAR QUARTER TRIM PANEL. FAILURE TO DO SO CAN RESULT IN UNEXPECTED INFLATION OR DEFLATION OF THE AIR SPRINGS, WHICH CAN RESULT IN SHIFTING OF THE VEHICLE DURING THESE OPERATIONS.

1. Position the vehicle on a hoist.
2. Remove the differential housing cover (3N136). Remove the bolts, differential housing cover and allow axle lube to drain.
3. Locate and remove the differential pinion shaft shaft retaining bolt (D8BZ-4241-B) as shown in Figures 1 and 2.
4. Remove the differential pinon shaft.

NOTE

IT IS ONLY NECESSARY TO PARTIALLY REMOVE THE PINION SHAFT ENOUGH TO

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.

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CLEAR THE SPACE BETWEEN THE PINION GEARS FOR S-SPRING INSERTION. IF YOU COMPLETELY REMOVE THE DIFFERENTIAL PINION SHAFT, BE CAREFUL NOT TO ALLOW THE PINION GEARS TO BECOME MISALIGNED.

5. Remove both axle shaft snap rings, as shown in Figure 3.
6. Clear and hold axle shafts beyond the face of the side gears.

NOTE
THE AXLE SHAFT SHOULD ONLY BE PULLED OUT FAR ENOUGH TO ALLOW INTRODUCTION OF THE S-SPRING. DO NOT PULL THE AXLE SHAFT PAST THE SIDE GEAR. ENSURE THAT THE AXLE SHAFT IS CONSTANTLY SUPPORTED IN TWO LOCATIONS, AT BOTH THE SHAFT BEARING AND THE SIDE GEAR. THIS WILL PREVENT THE AXLE SHAFT FROM DROPPING DOWN AT AN ANGLE AND DAMAGING THE OIL SEAL.

7. Place a differential clutch spring (EOAZ-4214-A) between the side gears of both RH and LH axleshafts, as shown in Figure 4. To install the differential clutch spring use a soft-faced hammer or hardwood block. Tap into position and center between side gears (Figure 4).
8. Push in on the axle shafts and install new snap rings N808765-S on RH and LH sides. Do not reuse the old snap rings.
9. Reinstall the differential pinion shaft and install a new pinion shaft retaining bolt, P/N D8BZ-4241-B.
10. Remove all traces of old silicone prior to reinstalling front differential housing cover, and clean metal mating surfaces with Metal Surface Cleaner, F4AZ-19A536-RA or equivalent meeting WSE-M5B392-A. Install the front differential cover using Silicone Gasket and Sealant F7AZ-19554-EA or equivalent meeting WSE-M4G323-A4. Following procedure in Section 205-03 of Workshop Manual, torque bolts to 24 Lb-ft (32 N•m).

NOTE
FRONT DIFFERENTIAL HOUSING COVER MUST BE INSTALLED WITHIN 15 MINUTES OF APPLYING SILICONE, OR SURFACES MUST BE RECLEANED AND SILICONE REAPPLIED. ALLOW SILICONE TO CURE ONE HOUR BEFORE FILLING.

11. Fill axle with 3.6 pints Motorcraft SAE 75W-90 Premium 4x4 Front Axle Lubricant XY-75W90-QFEHP. Lubricant level should be 0.38 in (9.56 mm) below bottom of differential housing fill hole.

PART NUMBER	PART NAME
EOAZ-4214-A	Differential Clutch Spring
F4AZ-19A536-RA	Metal Surface Cleaner
F7AZ-19554-EA	Silicone Gasket And Sealant
XY-75W90-QFEHP	Axle Lubricant
N808765-S	Snap Ring - Axle Shaft
D8BZ-4241-B	Pinion Shaft Retaining Bolt

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
022507A	Install S-Spring (Clutch Spring) Into The Front Axle Differential	1.7 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
3K147	42

OASIS CODES: 509000, 597997, 702000, 702100, 703000, 703200, 703300

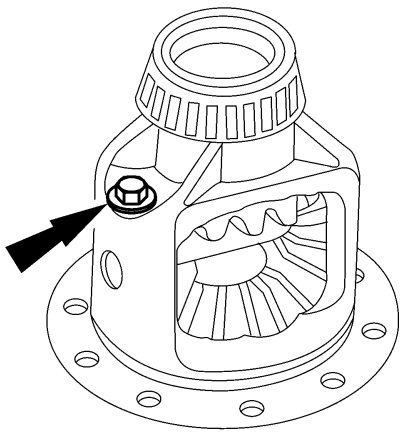


Figure 1 - Article 02-25-7

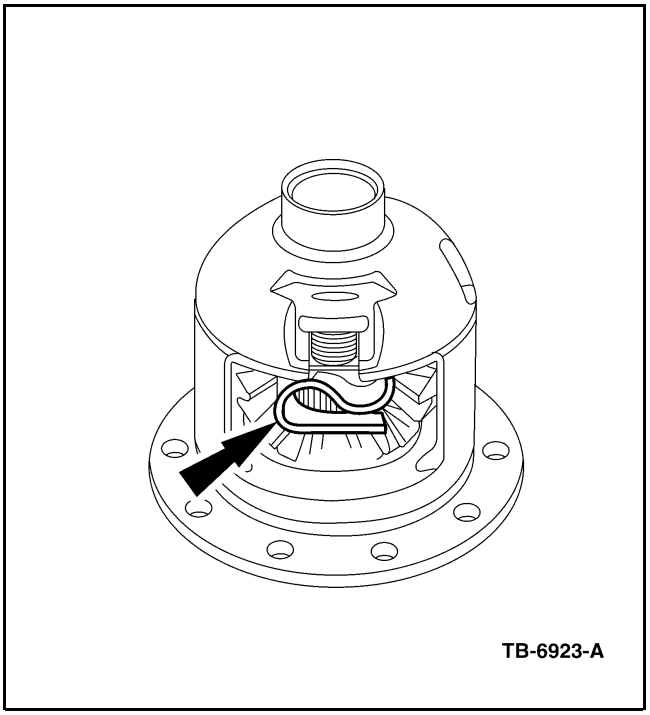


Figure 2 - Article 02-25-7

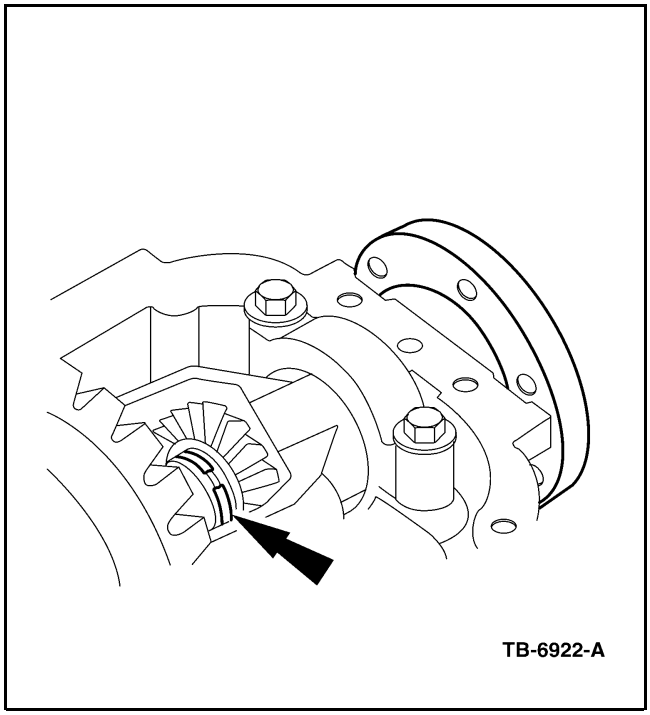


Figure 3 - Article 02-25-7

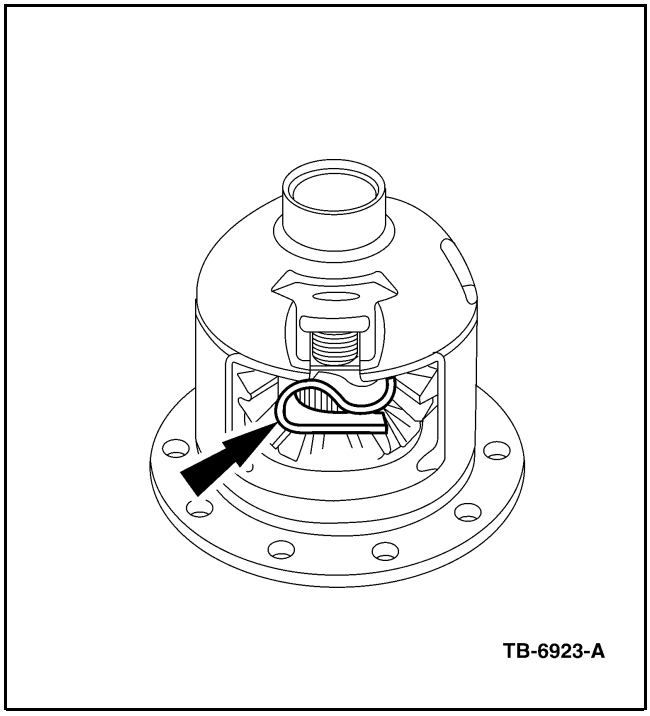


Figure 4 - Article 02-25-7